

Hyun-Sik Jeon

E-mail: hyunsik.jeons@gmail.com | Phone: (+82)10-7238-4009 | Website: <https://ddeonsik.github.io/>

Research Interest | *VLM, Autonomous Driving, 3D Scene Understanding, Motion Planning*

Education

Seoul National University of Science and Technology (SeoulTech)

Seoul, Republic of Korea

M.S. in Automotive Engineering

Mar 2024 – Feb 2026

- Thesis Advisor: *Prof. Jong-Eun Ha*
- Thesis title: Spatial Understanding for Camera-based Autonomous Driving Perception and 3D Visual Grounding
- GPA: 3.81 / 4.5 (3.39 / 4.0)

Shinhan University

Gyeonggi-do, Republic of Korea

B.S. in Future Automotive Engineering

Mar 2018 – Feb 2024

- GPA: 3.63 / 4.5 (3.23 / 4.0), Major GPA: 3.83 / 4.5 (3.40 / 4.0)

Publications and Presentations

1. (In Preparation) (Kor. / Eng. Abs.) **H.-S. Jeon**, J.-E. Ha*, “Enhancing Spatial Reasoning in VLMs through Saliency-Aware SpatialRGPT,” *Journal of Institute of Control, Robotics and Systems (ICROS)*, 2026.
2. (Submitted) S.-H. Kang, **H.-S. Jeon**, J.-E. Ha*, “Robust Zero-shot 3D Visual Grounding via Structural Query Decomposition and Dynamic Viewpoint Selection,” *Multimedia Tools and Applications (Springer)*, 2026.
3. **H.-S. Jeon**, S.-H. Kang, J.-E. Ha*, “Enhancing SeeGround with Relational Depth Text for 3D Visual Grounding,” *Applied Sciences (MDPI)*, vol. 16, no. 2, p. 652, 2026.
4. (Kor. / Eng. Abs.) **H.-S. Jeon**, Y.-M. Kim, J.-E. Ha*, “Object Mask Module for Enhancing Multi-view 3D Occupancy Perception Performance Based on ViewFormer,” *Journal of Institute of Control, Robotics and Systems (ICROS)*, vol. 31, no. 10, pp. 1160–1168, 2025.
5. (Kor. / Eng. Abs.) **H.-S. Jeon**, S.-M. Park, J.-E. Ha*, “UniAD Model Lightweighting and Performance Comparison,” *Journal of Institute of Control, Robotics and Systems (ICROS)*, vol. 31, no. 4, pp. 256–264, 2025.
6. (Kor.) J.-W. Lee, **H.-S. Jeon**, Y.-S. Choi*, “Steering and Velocity Control Between Traffic Cones for a LiDAR-based Autonomous Driving Platform in Low-spec Environments,” *Korea Institute of Information & Telecommunication Facilities Engineering (ITFE) Conference*, pp. 227–228, 2023.
7. (Kor.) J.-W. Lee, **H.-S. Jeon**, Y.-S. Choi*, “Using a LiDAR Sensor for Autonomous Driving in a Low-end Environment,” *The Institute of Electronics and Information Engineers (IEIE) Conference*, pp. 2434–2436, 2023.
8. (Kor.) S.-J. An, S.-T. Kim, **H.-S. Jeon**, Y.-S. Choi*, “A Method for Improving Lane Recognition by Adaptive Removing Vehicle Elements,” *Korea Institute of Information & Telecommunication Facilities Engineering (ITFE) Conference*, pp. 24–25, 2022.
9. (Kor.) **H.-S. Jeon**, S.-J. An, S.-T. Kim, Y.-S. Choi*, “Vehicle Element Removal Method to Improve Lane Recognition,” *The Institute of Electronics and Information Engineers (IEIE) Conference*, pp. 80–82, 2022.

Research Experiences

Artificial Intelligence System (AIS) Lab

Mar 2024 – Feb 2026

SeoulTech University, *Advisor: Prof. Jong-Eun Ha*

- Proposed Relational Depth Text for 3D Visual Grounding, improving accuracy by up to 6.74%.

Undergraduate Researcher

Jul 2022 – Dec 2023

Shinhan University, *Advisor: Prof. Yong-Soo Choi*

- Authored 4 conference papers on vision/LiDAR-based perception for low-spec autonomous platforms.

Projects

Dynamic Scene Understanding via Integration of Neural Rendering and Transformers (Ongoing Lab Project, PI: Prof. Ha) SeoulTech University, <i>Advisor: Prof. Jong-Eun Ha</i> Mid-Career Research Program, National Research Foundation of Korea (NRF), Ministry of Science and ICT (MSIT)	Mar 2024 – Feb 2026
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Professional Experiences

Vision Team Leader, Autonomous Driving Club “S.D.S” Shinhan University, <i>Advisor: Prof. Yong-Soo Choi</i> - Participant, 2023 University Creative Mobility Competition MOLIT, KATRI, KASA – Oct 2023 - Keywords: Vision-based Perception System, End-to-End Training Pipeline, Custom Traffic Dataset Curation	Mar 2022 – Feb 2024
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Certifications & Field Training

Ways1 Inc. — Field Practicum Development of E2E DL-based State Estimation Algorithm to Replace Kalman Filter	Jun 2024
NAVER Connect Foundation (Boostcourse) Neural Networks and Deep Learning	Oct 2023
Gyeonggi Jobs Foundation Electric Vehicle Maintenance Plus Course	Sep 2023
TÜV SÜD Korea EV High Voltage Safety Training Level 1	Aug 2023
HRD Korea (Human Resources Development Service of Korea) Industrial Engineer, Motor Vehicles Maintenance (National Qualification)	Jun 2022

Military Service (Mandatory)

Republic of Korea Air Force Sergeant, Vehicle Maintenance Specialist Completed	Feb 2020 – Nov 2021
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Skills

Domain Knowledge: 3D Visual Grounding, 3D Occupancy Prediction, BEV Perception, Multi-Modal, Model Lightweighting
ML/DL Frameworks: PyTorch, MMCV
Robotics & Sim: ROS2, IsaacGym
Programming: Python, MATLAB, LaTeX
Platforms & Environments: Linux (Ubuntu), Windows, Docker, Anaconda, Git/GitHub, Hugging Face
Mechanical Engineering Tools: CAD, CATIA, Abaqus, PAM-STAMP
Languages: Korean (Native), English (Proficient)